Side check: research results on data with NSE's own bars added

1. The 98-test scan

Done after the data check against NSE (section 7 of the data quality report), as a labelled side check. It is not the study itself: the submitted scan (`outputs/tables/anomaly_scan_all_tests.csv`), the grid and every number built on them are unchanged. Here the same code, the same seeds and the same cleaning run on the Yahoo research data with the real trading days that Yahoo lacks put back from NSE's files (the days listed in section 7 of the data quality report; Muhurat evenings left out). Research period only.

scan	cleaned days	tests	tradable tests	tradable with raw p<0.05	BH<0.05	Bonferroni<0.05
as submitted (Yahoo only)	2445	98	43	3	14	12
NSE days added	2459	98	43	3	14	12

Tests that survive the multiple-testing correction (BH 5%) in either scan:

family	test	index	effect sub / filled	t sub / filled	raw p sub / filled	BH p sub / filled	Bonferroni p sub / filled
volatility	Ljung-Box on squared returns (10 lags)	Nifty50	- / -	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
distribution	Jarque-Bera normality	BankNifty	12.51 / 12.6	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
distribution	Jarque-Bera normality	Nifty50	18.23 / 18.36	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
volatility	Ljung-Box on squared returns (10 lags)	BankNifty	- / -	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
volatility	ARCH-LM (5 lags)	Nifty50	424.16 / 426.85	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
volatility	ARCH-LM (5 lags)	BankNifty	277.07 / 279.5	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
gaps	average overnight (close to open) return	Nifty50	12.58 / 12.57	9.23 / 9.28	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
gaps	average overnight (close to open) return	BankNifty	12.88 / 12.75	7.38 / 7.36	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
random walk	Ljung-Box on returns (10 lags)	Nifty50	- / -	- / -	0.00000 / 0.00000	0.0000 / 0.0000	0.0000 / 0.0000
gaps	average intraday (open to close) return	Nifty50	-7.21 / -7.23	-4.35 / -4.37	0.00001 / 0.00001	0.0001 / 0.0001	0.0014 / 0.0012
gaps	yesterday's gap -> next tradable day	Nifty50	12.84 / 12.34	3.91 / 3.76	0.00009 / 0.00017	0.0008 / 0.0015	0.0091 / 0.0170
random walk	Ljung-Box on returns (10 lags)	BankNifty	- / -	- / -	0.00019 / 0.00019	0.0015 / 0.0016	0.0185 / 0.0189
random walk	Runs test (up/down closes)	Nifty50	-3.21 / -3.39	- / -	0.00132 / 0.00069	0.0100 / 0.0052	0.1297 / 0.0675
gaps	yesterday's gap -> next tradable day	BankNifty	18.9 / 18.37	3.05 / 2.95	0.00232 / 0.00316	0.0163 / 0.0221	0.2276 / 0.3094

Tests whose BH-significance changed: 0. Tests whose raw p-value crossed 0.05: 0. Largest change in a raw p-value across all 98 tests: 0.449 (ML walk-forward, gradient boosting (depth 2) direction accuracy 2016-2021, BankNifty).

The eight largest changes in raw p-value:

family	test	index	raw p sub	raw p filled	BH p sub	BH p filled
ML walk-forward	gradient boosting (depth 2) direction accuracy 2016-2021	BankNifty	0.9598	0.5105	0.998	0.922
ML walk-forward	logistic (L2) direction accuracy 2016-2021	BankNifty	0.7596	0.5928	0.989	0.922
streaks	after 3+ down closes -> next tradable day	Nifty50	0.3778	0.2275	0.904	0.758
expiry	day after expiry close-to-close	BankNifty	0.6941	0.5590	0.986	0.922
expiry	day after expiry open-to-open	BankNifty	0.7519	0.6339	0.989	0.922
streaks	after 4+ up closes -> next tradable day	BankNifty	0.4753	0.5693	0.904	0.922
day of week	Wed close-to-close vs other days	Nifty50	0.7727	0.8666	0.989	0.998
ML outliers	after isolation-forest outlier day -> next tradable Nifty	Nifty50	0.9071	0.9979	0.998	0.998

2. The grid, the choice of rule and the reality check

Same grid of 18 variants, the same selection rule (written before the run) and the same bootstrap seed, on the research data plus NSE's bars. Nothing in `grid_research.csv` or `selected_params.json` is touched.

k	hold	side	net Sharpe sub / filled	Sharpe 12-16 sub / filled	Sharpe 17-21 sub / filled	trades sub / filled
0.5	1	both	0.968 / 0.938	1.16 / 1.09	0.82 / 0.83	860 / 863
0.5	1	long	0.955 / 0.955	1.19 / 1.16	0.78 / 0.81	563 / 564
0.5	2	both	1.246 / 1.227	1.21 / 1.23	1.28 / 1.24	661 / 663
0.5	2	long	1.169 / 1.179	1.19 / 1.23	1.16 / 1.15	385 / 385
0.5	3	both	1.086 / 1.084	1.11 / 1.14	1.07 / 1.04	578 / 584
0.5	3	long	0.994 / 0.985	0.97 / 0.97	1.02 / 1.00	318 / 320
1.0	1	both	1.069 / 1.048	1.21 / 1.15	0.98 / 0.99	511 / 514
1.0	1	long	1.081 / 1.102	1.03 / 1.08	1.13 / 1.13	340 / 342
1.0	2	both	1.130 / 1.118	1.01 / 0.96	1.24 / 1.26	430 / 429
1.0	2	long	1.159 / 1.162	1.10 / 1.10	1.22 / 1.22	273 / 272
1.0	3	both	0.899 / 0.881	0.94 / 0.89	0.87 / 0.89	381 / 381
1.0	3	long	0.991 / 0.982	1.21 / 1.17	0.83 / 0.84	233 / 233
1.5	1	both	0.656 / 0.550	0.74 / 0.47	0.62 / 0.62	269 / 269
1.5	1	long	0.759 / 0.715	0.57 / 0.45	0.91 / 0.92	167 / 169
1.5	2	both	0.998 / 0.968	0.98 / 0.88	1.03 / 1.06	244 / 243
1.5	2	long	1.025 / 0.982	1.19 / 1.06	0.90 / 0.93	147 / 148
1.5	3	both	0.875 / 0.843	1.08 / 0.96	0.75 / 0.78	223 / 222
1.5	3	long	0.903 / 0.864	1.16 / 1.05	0.72 / 0.73	132 / 134

Chosen variant: as submitted k=0.5, hold=2, side=both; with NSE's bars added k=0.5, hold=2, side=both (the same variant). **Reality check p-value:** 0.003 as submitted, 0.004 with NSE's bars added. Largest change in a variant's net Sharpe: 0.106.